

The question for every engineering team: If 65% of your PRs ship with no real review, and AI is generating code 1.7x more likely to have defects, what is the cost of doing nothing?

Default Inputs (50-Developer Team)

INPUT	DEFAULT VALUE
Number of developers	50
Avg fully-loaded annual compensation	\$175,000
PRs per developer per week	4
% of PRs that get meaningful review today	35%
Using AI coding assistants?	Yes
Production incidents per quarter	4

1. Rubber-Stamp Cost: Unreviewed PRs

65% of merged PRs are approved after minimal review -- zero comments or fewer than 3 words total (Graphite, 2025 study of 50M+ PRs).

Unreviewed PRs/year = 50 devs x 4 PRs/wk x 52 wks x 0.65 = 6,760
Defects (1 per 10 unreviewed) = 676/year
Cost to fix in production (30x multiplier, NIST) = \$4,500 each

Annual cost of escaped defects = 676 x \$4,500 = \$3,042,000

With AI Coding Assistants (the Multiplier)

AI-generated code: 1.7x more issues (CodeRabbit, 470-PR study), 2.74x more security vulns (Veracode), 322% increase in privilege escalation (Apiiro).

AI-adjusted defects = 676 x 1.7 = 1,149/year
AI-adjusted cost = 1,149 x \$4,500 = \$5,170,500/year

Security incidents (1% of 1,852 security issues)
= 18 x \$150,000 = \$2,700,000

2. Production Incident Cost

Annual = 16 incidents x 8 hrs x 3 people x \$84/hr = \$32,256 direct
Fully loaded (7x multiplier) = \$225,792/year

3. Manual Review Time Tax

Current review labor = $50 \times 4 \times 1.5 \text{ hrs} \times 52 \times \$84/\text{hr} = \$1,310,400/\text{year}$
GuardSpine saves 40–60% (automates L0–L2) = $\$655,200/\text{year}$ recovered

4. Technical Debt Prevention

Debt labor = $50 \text{ devs} \times 13.5 \text{ hrs/wk} \times 52 \text{ wks} \times \$84/\text{hr} = \$2,948,400/\text{year}$
GuardSpine prevents 15–25% of new debt = $\$589,680/\text{year}$ avoided

Total Annual Cost Without GuardSpine

RISK CATEGORY	ANNUAL COST
Escaped defects from rubber-stamped PRs	\$3,042,000
AI code defect multiplier	+\$2,128,500
Production incident costs (fully loaded)	\$225,792
Code review labor (opportunity cost)	\$1,310,400
Technical debt accumulation	\$589,680
Total annual risk exposure	\$7,296,372

ROI with GuardSpine (Team Tier: \$19,200/year)

\$3.1M

ANNUAL VALUE (CONSERVATIVE)

16,337%

ROI

~2 days

PAYBACK PERIOD

VALUE DRIVER	% CAPTURED	ANNUAL VALUE
Defects caught pre-merge	40%	\$1,216,800
AI defect multiplier reduction	30%	\$638,550
Incidents prevented	25%	\$56,448
Review time recovered	50%	\$655,200
Tech debt avoided	20%	\$589,680
Total annual value		\$3,156,678
GuardSpine Team cost		(\$19,200)
Net savings		\$3,137,478

Even at 10% of these estimates, the ROI is 1,633% -- \$315K savings on a \$19K investment.

Benchmarks Cited

CLAIM	SOURCE	YEAR
65% of PRs approved with minimal/zero review	Graphite (50M+ PRs)	2025
30x cost multiplier: production vs review	NIST, HackerOne	2024
1.7x more issues in AI-generated code	CodeRabbit (470 PRs)	2024
2.74x more security issues in AI code	Veracode (100+ LLMs)	2025
322% increase in privilege escalation	Apiiro	2025
\$4.44M avg breach cost (global)	IBM Cost of a Data Breach	2025
13.5 hrs/week on tech debt per dev	Stripe/HackerOne	2024

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